

Active Directory GPO Implementation Lab (Windows Server 2025)

Project Overview

This lab project demonstrates how to centrally manage, restrict, and configure client workstations in an enterprise network using **Group Policy Objects (GPOs)**. Operating in a **Windows Server 2025** environment, I built a domain infrastructure, created targeted Organizational Units (OUs), and deployed multiple GPOs to enforce corporate wallpaper standards, restrict access to sensitive system settings, configure logon banners, and automatically map network drives.

Topology & Lab Environment

- **Domain Controller VM:** Windows Server 2025 (Hostname: SERVER0 | Domain: college.com)
- **Client Workstation VM:** Windows 10 Enterprise (Testing with domain user account: jeran rai)
- **Network Setup:** Dedicated Layer 2 host-only network inside VMware Workstation Pro ensuring isolated, secure communication.

Step-by-Step Implementation

Step 1: Active Directory OU Setup

- Opened **Active Directory Users and Computers (ADUC)** on SERVER0.
- Created a new Organizational Unit (OU) named Computer Science.
- Provisioned the test user account jeran rai and added the Windows 10 computer object directly into the Computer Science OU to ensure targeted policy application.

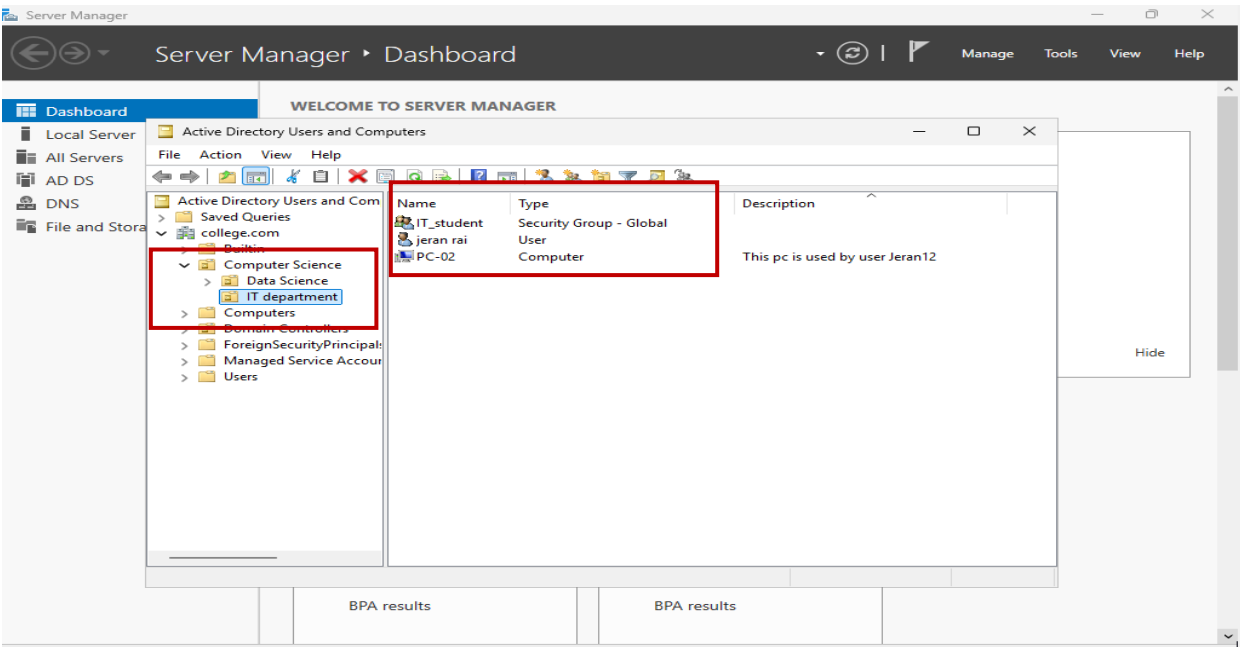


Fig.No:1 Users in OU Computer Science

Step 2: Preparing the Centralized Storage (UNC Share)

- To deploy the corporate wallpaper via GPO, the file had to be accessible over the network:
- Created a local folder on the drive **C:\wallpaper\image2.jpg** inside it.

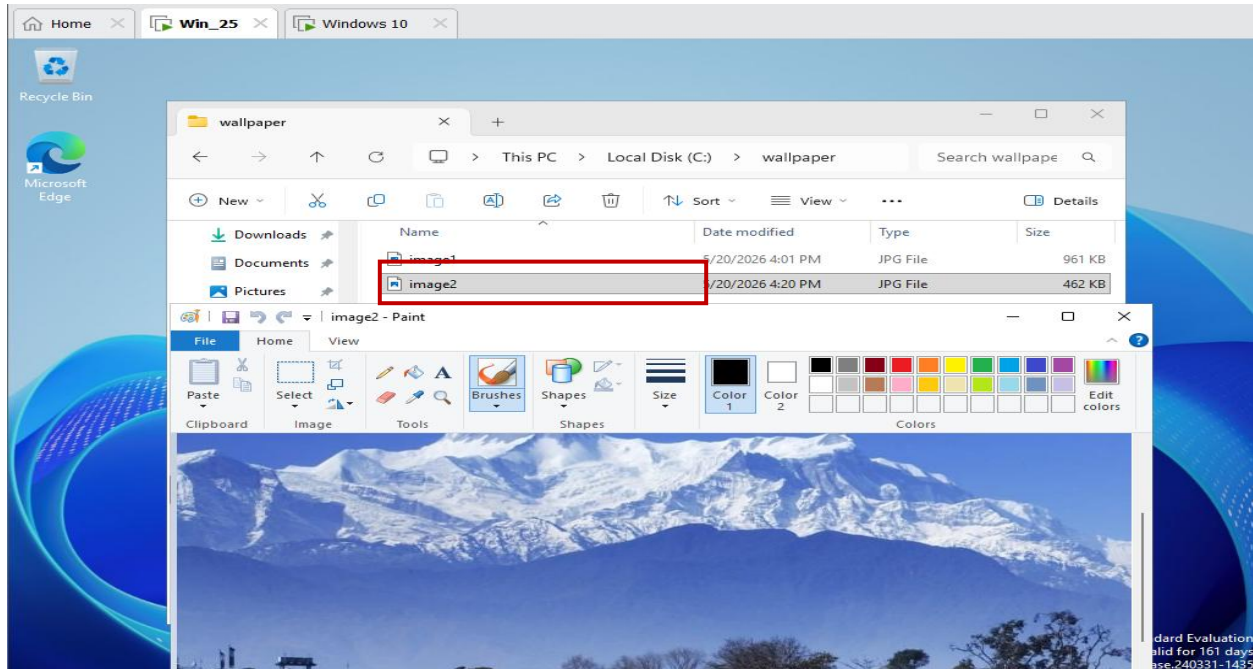
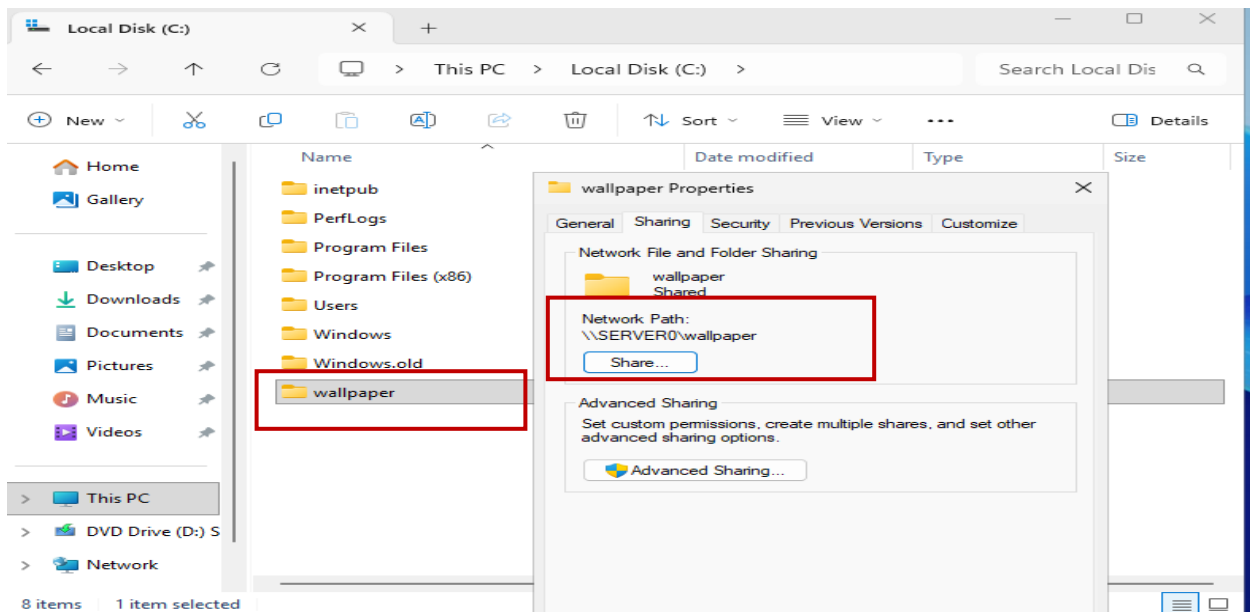


Fig No.: 2 Sharing Folder

- Configured **Advanced Sharing** to share the folder as wallpaper, making the network path <\\SERVER0\wallpaper\image2.jpg>.



- Adjusted **Share Permissions** to grant *Everyone* Read access. If access given to user_name, then only access to that user. Other user can't access read permission for UNC path and GPO will fail. However, everyone read permission get all other users read authorization.
- Adjusted **NTFS Security Permissions** to ensure Domain Users/Computers could explicitly read and execute the file.

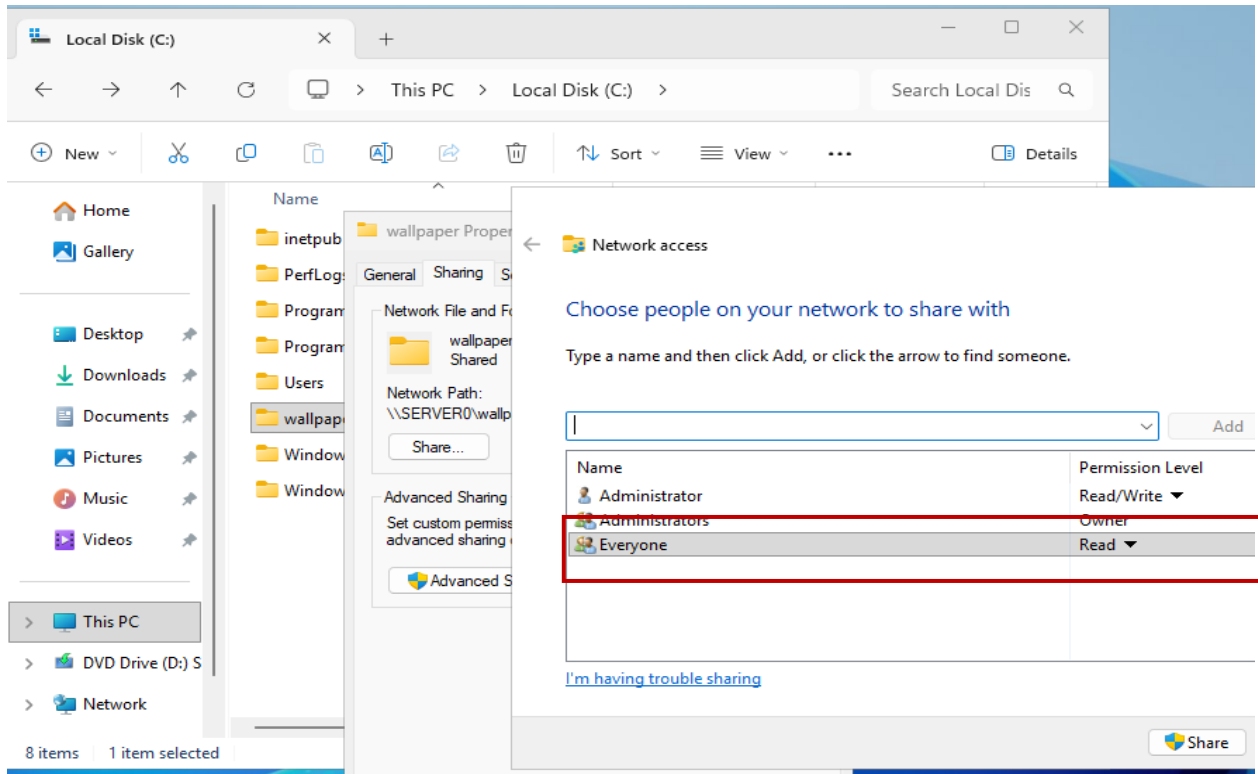


Fig No.: 3 Granting person to Everyone

Step 3: Configuring Group Policies

Opened the **Group Policy Management Console (GPMC)** and created/linked a series of policies on the Computer Science OU:

3.1 Wallpaper Deployment (wallpaper policy):

- Navigated to: **User Configuration -> Administrative Templates -> Desktop -> Desktop -> Desktop Wallpaper**.
- Enabled the policy and set the wallpaper path to the UNC format: `\\SERVER0\wallpaper\image2.jpg`. Set the wallpaper style to **Fill**.

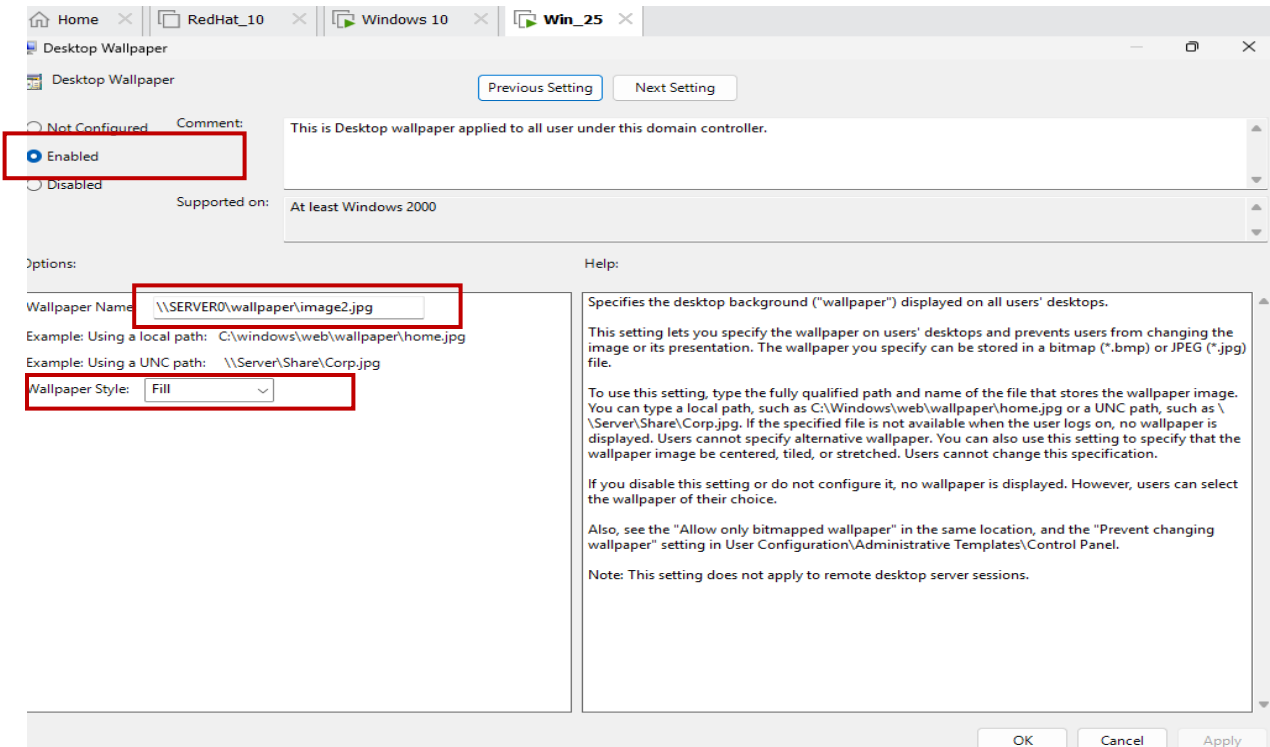


Fig No.: 4 Wallpaper Deployment

3.2 System Hardening (control panel restriction):

- Configured user restrictions to prevent standard users from tampering with local network adapters or control panel restriction.
- Navigation Path: **User Configuration → Policies → Administrative Templates → Control Panel → Prohibit access to Control Panel and PC settings**

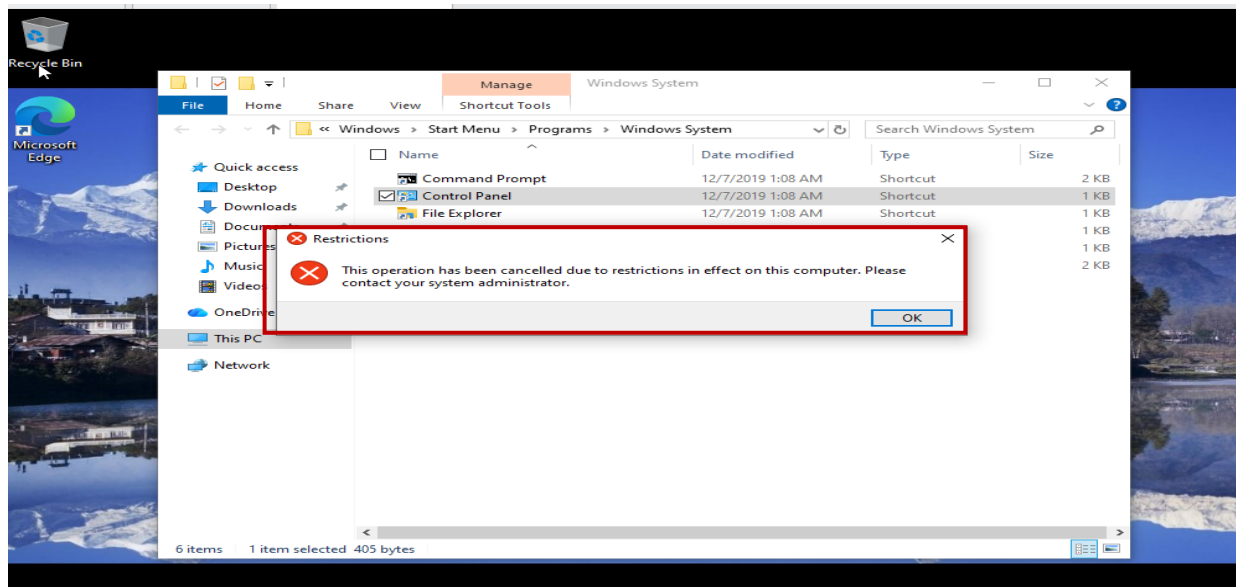


Fig No.: 5 Disable Control Panel

- Navigation Path: **User Configuration → Policies → Administrative Templates → Network → Network Connections → Prohibit access to properties of a LAN connection**

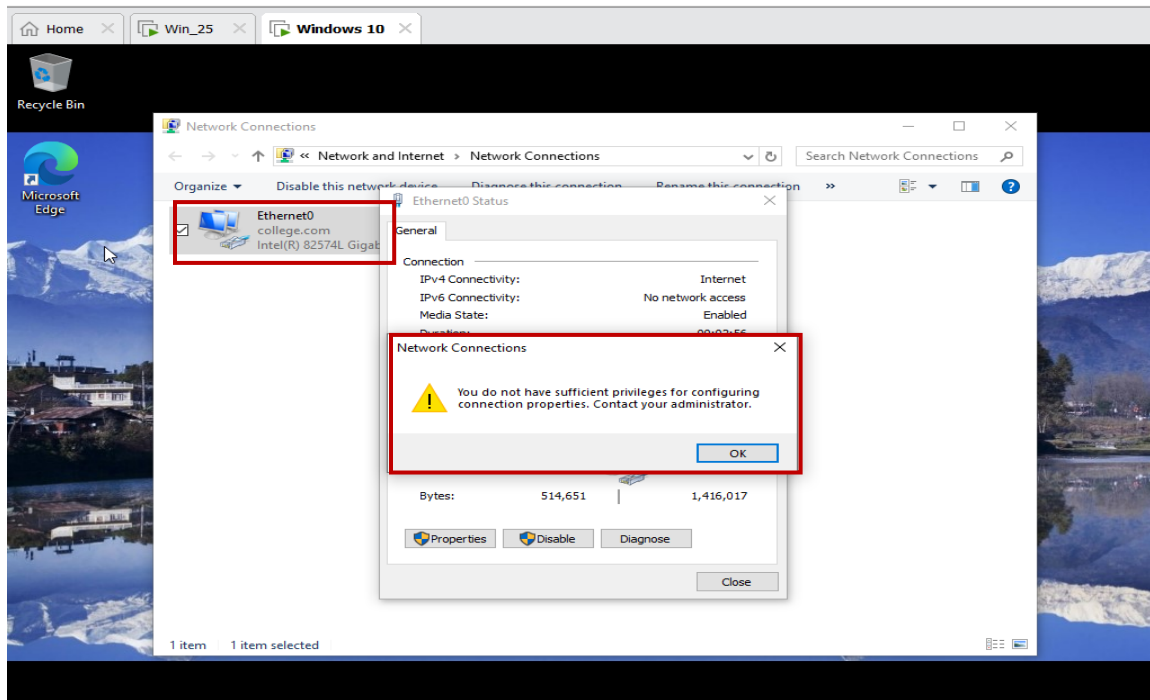


Fig No.: 6 Disable Network Adapter

3.3 Network Drive Mapping (map network drive):

- Utilized Group Policy Preferences (GPP) to map corporate file shares to client machines automatically upon login, removing the need for legacy batch login scripts.
- Navigation Path: **User Configuration → Preferences → Windows Settings → Drive Maps.**

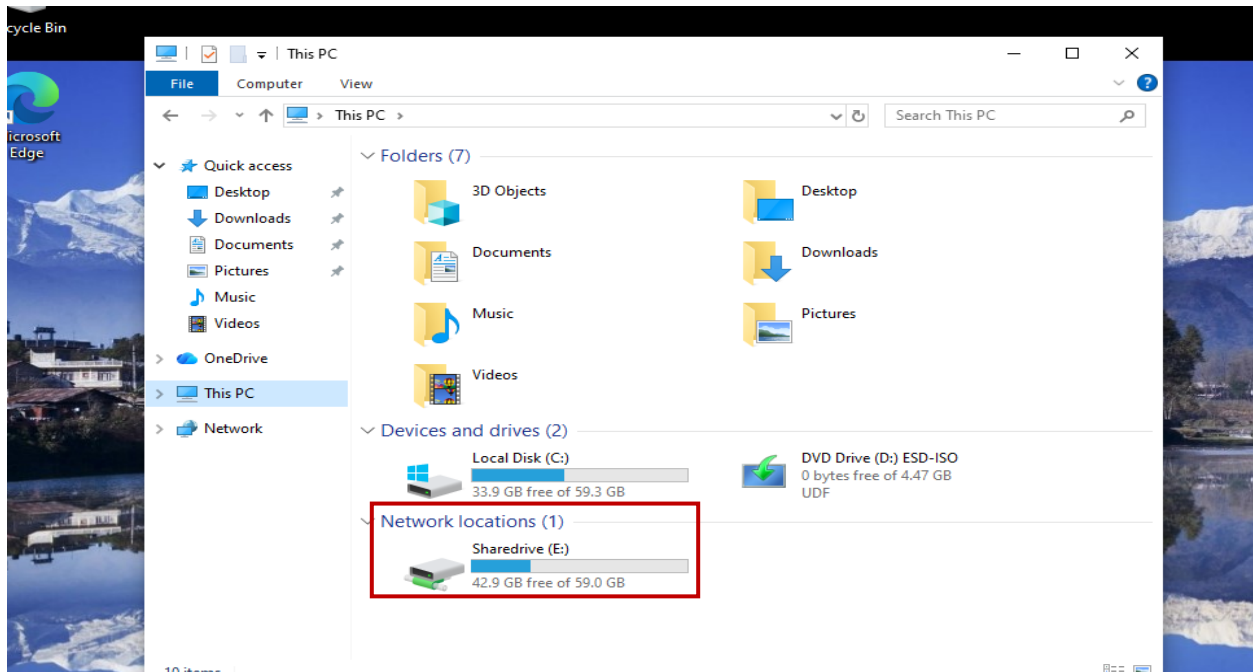
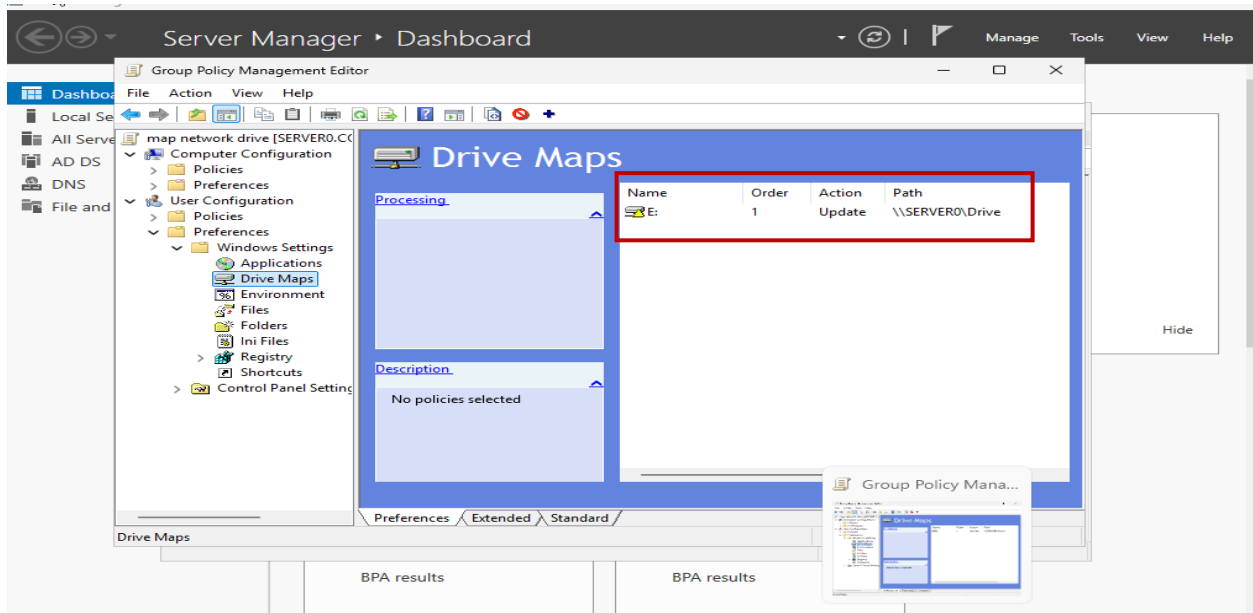


Fig No.: 7 Network Drive Mapping

3.4 Corporate Compliance (banner logon):

- Enabled interactive logon messages forcing an organization warning to display before a user can sign in.
- Navigation Path: **Computer Configuration → Policies → Windows Settings → Security Settings → Local Policies → Security Options → Interactive logon: Message title for users attempting to log on.**

- Navigation Path: **Computer Configuration → Policies → Windows Settings → Security Settings → Local Policies → Security Options → Interactive logon: Message text for users attempting to log on.**

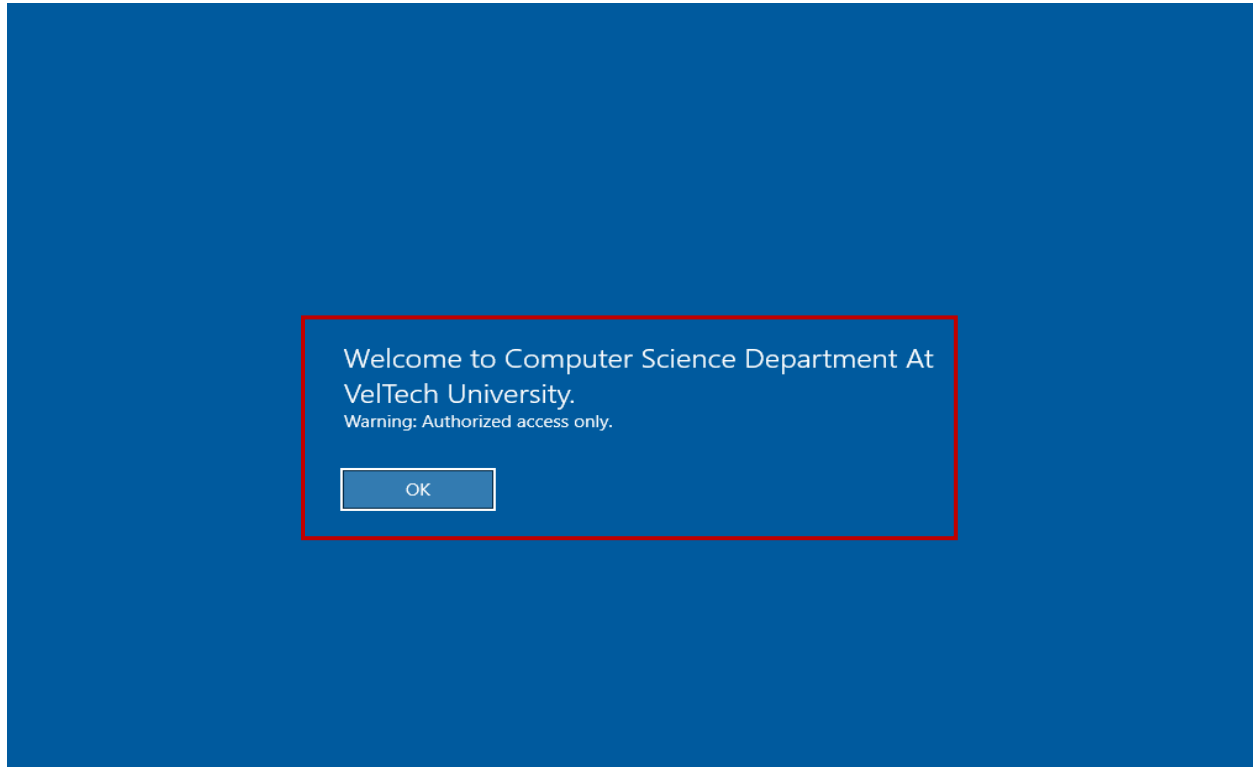
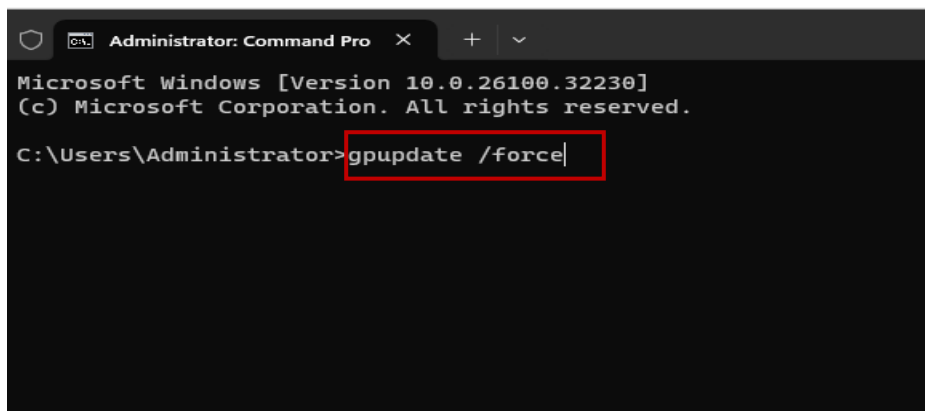


Fig No.: 8 Logon Banner

Step 4: Verification & Troubleshooting

Logged into the Windows 10 client machine as college\jieran rai.

Opened the command prompt and ran a forced policy update to pull the new changes from the Server 2025 Domain Controller immediately:



Confirmed that the corporate desktop wallpaper loaded successfully from the network path, the interactive login banner popped up, and unauthorized Control Panel options were successfully locked out.

Lessons Learned & Troubleshooting Insights

The Importance of UNC Paths: Initially learned that using a local path (like C:\images\photo.jpg) in a GPO fails on client machines because they look for that folder locally on their own C: drive. Using the network share path (\\SERVER0\wallpaper\) is mandatory for successful deployment.

Share vs. NTFS Permissions: When a client accesses a file over the network in Windows Server, both **share permissions** and **NTFS permissions** are evaluated together, and access is granted only if **both allow it**, with the **most restrictive permission winning**. Share permissions act as the first gate at the network level (when using a UNC path like \\Server\Share), while NTFS permissions control the actual access to the file or folder on the disk. Even if one layer allows full access, the other can still restrict or completely block it. For example, if the share permission allows “Full Control” but NTFS only allows “Read,” the user will only get read access.

In scenarios like applying a wallpaper via Group Policy, this becomes critical because the request is often made by the **computer account (SYSTEM context)** rather than the user. If either the share or NTFS permissions do not include the appropriate user or computer account (such as “Authenticated Users” or “Domain Computers”), the client will fail to access the file, and the wallpaper will not apply. Therefore, both permission layers must be properly aligned to ensure successful access.